

ACFI315 Finance with Excel

Lecture 6 Asset Pricing Models

Dr Xi Fu



Capital Market Line (CML)

- Investors optimize portfolios based on the mean-variance framework.
- Investors obtain the efficient frontier using standard deviation and expected return of all possible portfolios.
- The optimal capital allocation line is the tangent line to the efficient frontier through the risk-free rate.
- Market portfolio is aggregation of all risky portfolios.

Capital Market Line (CML)

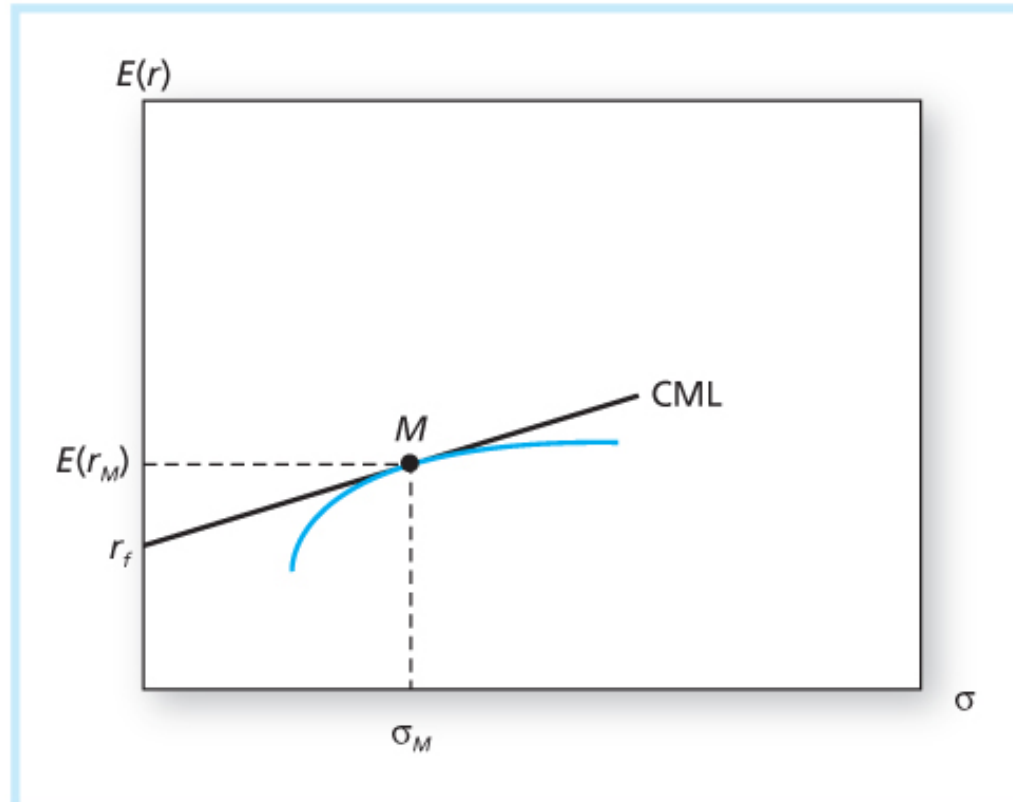


Figure 9.1 The efficient frontier and the capital market line

Sharpe Ratio

- The Sharpe Ratio measures how much excess return an investor receives for the extra volatility for holding a riskier asset.

$$\text{Sharpe Ratio} = \frac{E(R_p - R_f)}{\sigma_P}$$

- The Sharpe Ratio is the slope of the Capital Market Line.

The Capital Asset Pricing Model (CAPM)

- Based on the mean-variance framework, Lintner (1965) and Mossin (1966) find a linear relationship between the expected return and systematic risk. This relationship is later on acknowledged as the Capital Asset Pricing Model (CAPM).

$$E[r_i] = r_f + \beta_i(E[r_M] - r_f)$$

- The CAPM focuses on the systematic variance risk, which is captured by β .

$$\hat{\beta}_i = \frac{Cov(r_M, r_i)}{\sigma_M^2}$$

Security Market Line (SML)

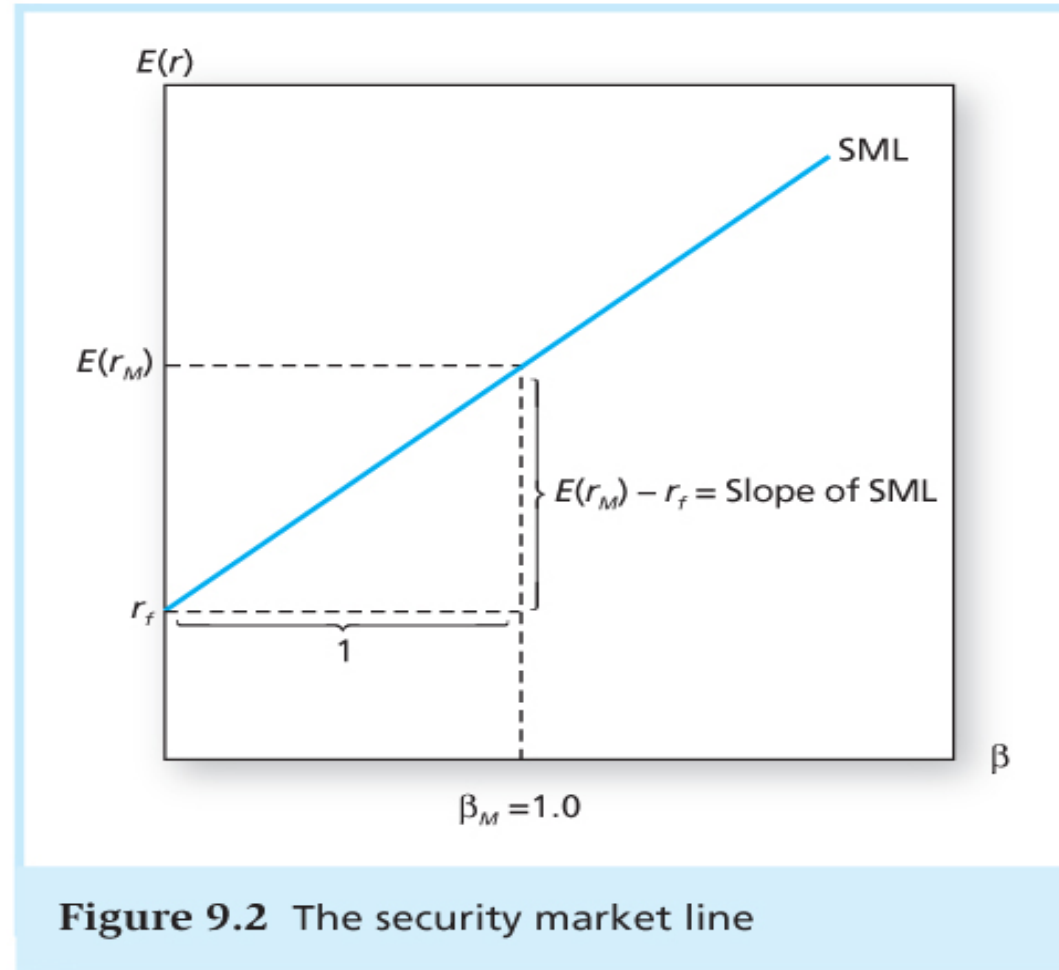


Figure 9.2 The security market line

Treynor Ratio

- The Treynor Ratio, also known as the reward-to-volatility ratio, measures how much excess return an investor receives for each unit of risk of a portfolio. Risk in the Treynor ratio refers to the systematic risk as measured by a portfolio's beta.

$$\text{Treynor Ratio} = \frac{E(R_p - R_f)}{\beta_P}$$

- The Treynor Ratio is the slope of the Security Market Line.

α and Investment Performance

- We could use the time-series data to estimate the CAPM:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i(r_{M,t} - r_{f,t}) + e_{i,t}$$

- The α_i included in the CAPM is of particular interest to investors and fund managers, in spite of the fact that the model assumes that $\alpha_i = 0$.
- It captures the additional return earned over and above the level implied by the β and the market risk premium.
- As such, it is considered a measure of risk-adjusted return, i.e., a rate of return that takes into account the investment's risk relative to that of the market portfolio,
- A positive α is often viewed as a gauge of a fund manager's skill in identifying profitable investment opportunities.

Comments on the CAPM

- Empirical evidence shows that the CAPM fail to adequately describe behaviours of stock returns:
 - Most of the assumptions of the CAPM conflict with the capital market (e.g. no transaction costs, no taxes, no restrictions on short sale, and etc.).
 - We cannot observe all tradable assets.
 - The CAPM focuses on the systematic risk only.

The Arbitrage Pricing Theory

- Ross (1976) proposes a multifactor model, the Arbitrage Pricing Theory (APT). The APT claims that the expected return of an asset should be a linear function of the asset's sensitivities to many different risk factors.

$$E(r_i) = r_f + \sum_{j=1}^J \beta_{ij} \lambda_j$$

- The CAPM is a special case of the APT.

Other Famous Multifactor Models

- Fama-French Three-Factor Model:

$$r_i = r_f + \beta_i(r_M - r_f) + s_iSMB + h_iHML$$

- Carhart Four-Factor Model:

$$r_i = r_f + \beta_i(r_M - r_f) + s_iSMB + h_iHML + m_iUMD$$

- Kraus and Litzenberger Three-Moment Model:

$$E(r_i) = r_f + b_1\beta_i + b_2\gamma_i$$